

National University
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Final Report

Embedded Control for Robotics

Semester Project

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Topic: Robot Operating System 2 (ROS-II)

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Abstract

Robot Operating System 2 (ROS 2) is a next-generation, open-source robotics middleware framework designed to address the performance, reliability, and scalability limitations of ROS 1. Built upon the Data Distribution Service (DDS) standard, ROS 2 enables real-time communication, enhanced security, multi-robot coordination, and cross-platform deployment. This paper provides an in-depth analysis of the architectural innovations, communication primitives, and industrial applications of ROS 2. Additionally, it reviews its evolution from ROS 1, explores its integration with simulation tools, highlights its security and real-time capabilities, and outlines the challenges that remain. The findings demonstrate that ROS 2 significantly advances robotics middleware for modern autonomous systems, offering a flexible and robust foundation for research, industry, and future robotics innovation.

Introduction

Robotics systems have become increasingly complex, requiring advanced software frameworks that support distributed computing, real-time processing, and interoperability between heterogeneous devices.

To tackle all the challenges in robotic systems, ROS (Robot Operating System) was developed which is more than a decade old open-source robotics middleware software, initially developed by two PhD students from Stanford University.

However, ROS 1 was not designed with determinism, security, or large-scale industrial deployment as primary goals. As robotics transitioned into commercial domains such as manufacturing, healthcare, logistics, and autonomous vehicles, a more robust middleware was needed. ROS 2 was introduced in 2017 to address these limitations by leveraging modern communication standards and improved architectural principles.

Fast-forward to 2024, ROS has evolved into a rich ecosystem of utilities, algorithms, and sample applications, transcending its origins as middleware software, and is now used by millions of people and thousands of companies worldwide.



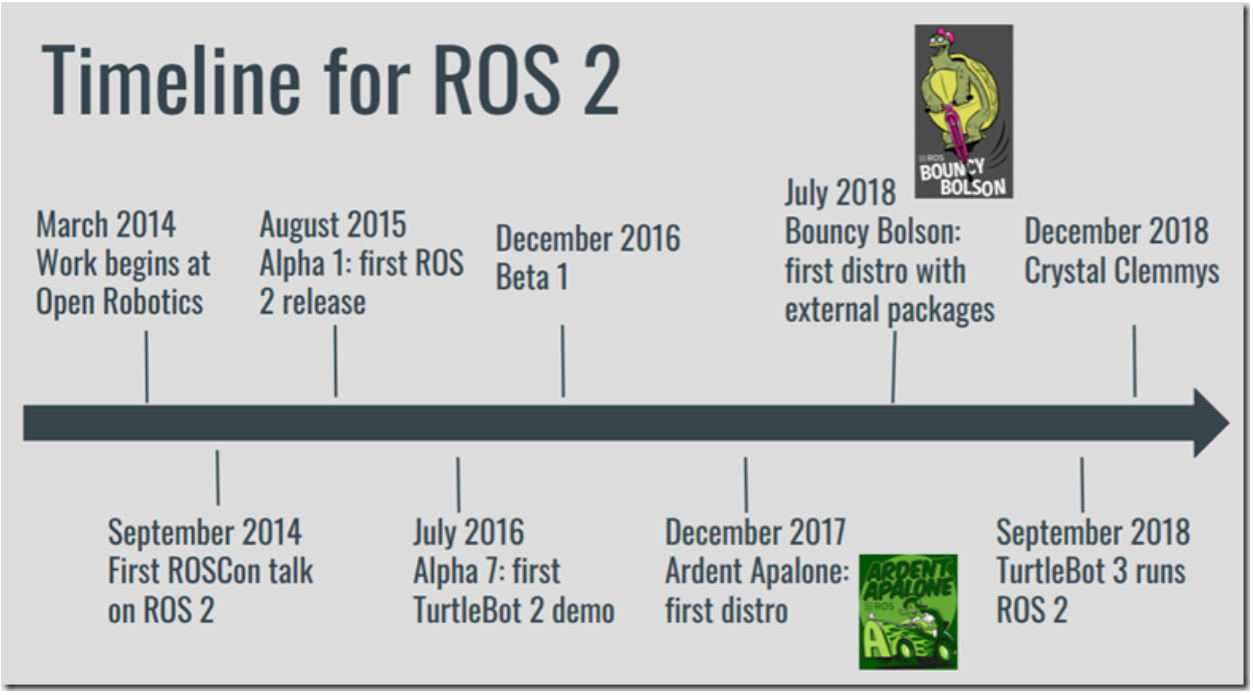
Evolution From ROS-I to ROS-II

ROS 1 provided a flexible framework but lacked critical features needed for robust robotic systems. Its message-passing system relied on a centralized master node (roscore), which created a single point of failure and limited scalability. Furthermore, ROS 1 did not support real-time execution or built-in security.

ROS 2 rearchitected these core components. Instead of a centralized master, ROS 2 employs a fully distributed discovery mechanism using DDS, enabling dynamic peer-to-peer communication. Other key improvements include:

- **Real-time support:** Deterministic communication through DDS QoS profiles.
- **Security:** ROS 2 includes Secure ROS 2 (SROS2), providing authentication, encryption, and access control.
- **Cross-platform compatibility:** Linux, Windows, macOS, and embedded systems such as microcontrollers.
- **Lifecycle nodes:** Enabling deterministic state management.
- **Improved package management:** Introduced the colcon build system.

These enhancements make ROS 2 more suitable for commercial robotics applications requiring reliability and long-term deployment.



ROS-II Architecture

ROS 2's architecture is modular and built around the DDS middleware. Its key components include nodes, topics, services, actions, and parameters, all of which support distributed robotic systems.

DDS Middleware

DDS is a publish–subscribe communication standard used in mission-critical systems such as aerospace and defense. In ROS 2, DDS provides:

- **Quality of Service (QoS):** Developers can configure reliability, durability, latency, and delivery deadlines.
- **Decentralized communication:** No master node required.
- **Real-time messaging:** Suitable for robotic control loops.

The integration of DDS makes ROS 2 significantly more robust and scalable compared to ROS 1.

Nodes

Nodes are independent processes that perform specific tasks such as sensor reading, motor control, or decision making. ROS 2 nodes are designed to be lightweight and can be distributed across multiple machines.

Topics

Topics provide asynchronous publish–subscribe messaging. This model is ideal for continuous data streams like sensor readings. ROS 2's QoS configuration ensures message delivery even under network constraints.

Services

Services enable synchronous request–response communication, suitable for tasks requiring immediate feedback, such as sending commands.

Actions

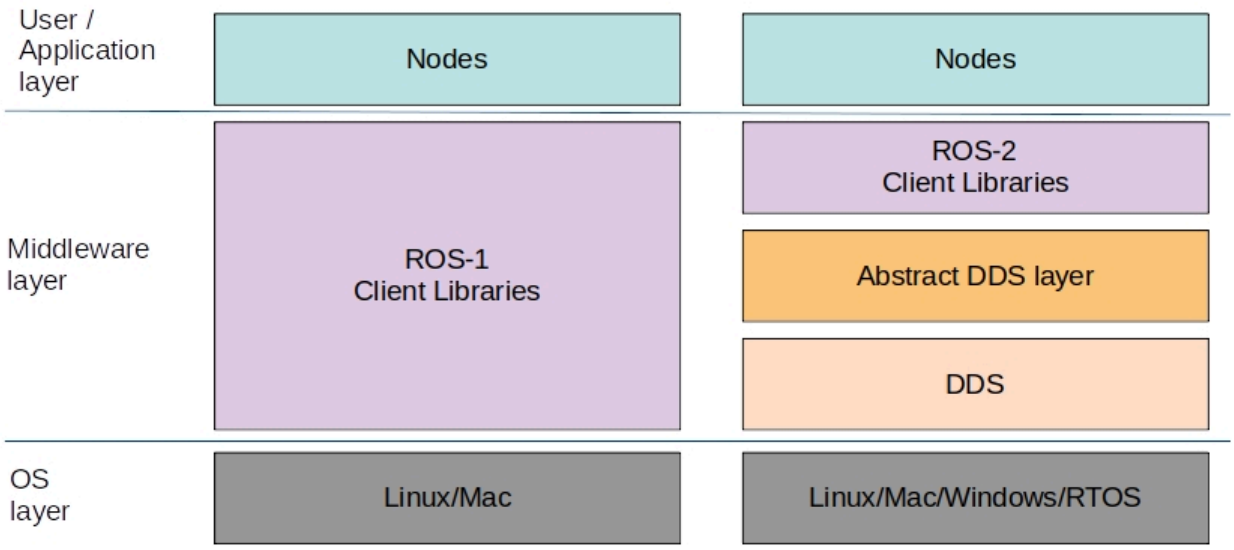
Actions extend the service model for long-running tasks (e.g., navigation goals). They support preemption, feedback, and cancellation.

Parameters

Parameters allow configurable settings within nodes and can be dynamically modified at runtime.

Lifecycle Nodes

Lifecycle nodes introduce deterministic state transitions—unconfigured, inactive, active, and finalized—making ROS 2 suitable for safety-critical systems.



Simulation and Visualization Tools

Simulation is essential for developing and testing robotic systems. ROS 2 supports several tools:

- **Gazebo (Ignition Gazebo):** Enables physics-based simulation of robots and environments.
- **RViz2:** A visualization tool for sensor data, coordinate frames, and robot states.
- **rqt:** Modular GUI tools for debugging and monitoring.
- **Foxglove Studio:** A modern visualization and debugging platform for ROS 2 data.

These tools allow testing algorithms in virtual environments before deployment on physical robots, reducing costs and risk.

Security: SROS2

ROS 1 lacked native security features, making it unsuitable for commercial applications. ROS 2 incorporates Secure ROS 2 (SROS2), which includes:

- **Authentication:** Validates nodes and users.
- **Encryption:** Secures communication channels.
- **Access control:** Defines permissions for individual nodes and resources.

SROS2 follows industry standards such as PKI (Public Key Infrastructure) and enhances the safety of networked robotic systems.

Compatibility of ROS-II with Microcontrollers

Direct deployment of full ROS 2 on microcontrollers is generally infeasible because ROS 2 (as implemented for desktop and embedded Linux) expects more CPU, memory, and OS services than are available on typical microcontrollers.

To bridge this gap, the ROS community developed micro-ROS, a purpose-built adaptation of ROS 2 for resource-constrained devices.

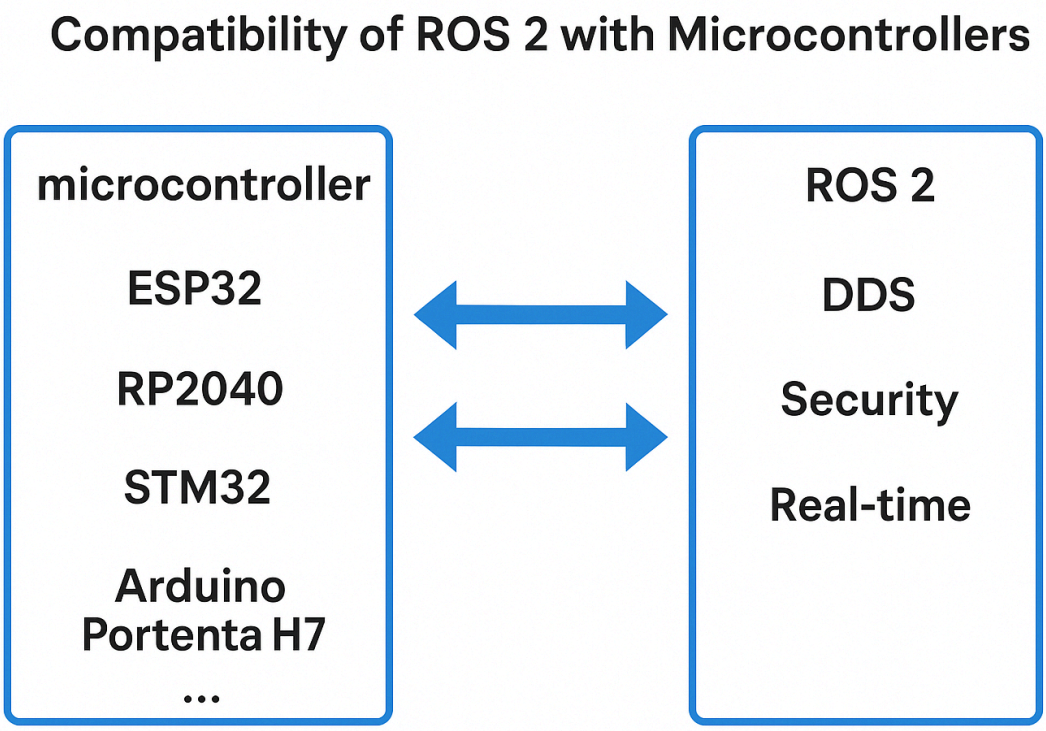
Supported RTOSes and Middleware

micro-ROS supports several real-time operating systems (RTOS) commonly used in embedded development notably FreeRTOS, Zephyr, and NuttX, and can also be used on Linux-based embedded boards.

Representative Microcontrollers and Boards (Support Status)

The micro-ROS project and related ecosystem repositories maintain an evolving list of officially supported and community-supported boards. Examples of widely supported microcontrollers/boards (as of the latest documentation) include:

- Espressif ESP32
- Raspberry Pi Pico (RP2040)
- Arduino Portenta H7
- STM32 family / ST Nucleo / STM32-based OpenCR
- Renesas RA6M5 and RX65N
- Teensy (selected models)



Applications of ROS 2

ROS 2 is widely used across industries:

1) **Autonomous Vehicles**

ROS 2 is widely adopted in the development of autonomous ground vehicles due to its integration of perception, localization, mapping, planning, and control modules. Many autonomous driving stacks, such as Autoware.Auto, rely on ROS 2 to handle large volumes of sensor data from LiDAR, radar, and cameras.

2) **Drones and Unmanned Aerial Vehicles (UAVs)**

The UAV industry benefits from ROS 2's lightweight communication and support for multi-robot systems. Drones often operate under constrained computing environments and rely on distributed decision-making when performing tasks such as aerial mapping, agriculture monitoring, infrastructure inspection, and search-and-rescue operations.

3) Industrial Automation and Manufacturing

Modern manufacturing environments increasingly deploy autonomous mobile robots (AMRs), robotic arms, and collaborative robots (cobots). Robotics companies employ ROS 2 to integrate sensor fusion, path planning, manipulation, and human–robot interaction. ROS 2's deterministic communication and lifecycle management make it well-suited for industrial applications

4) Healthcare and Assistive Robotics

ROS 2 has significant applications in healthcare robotics due to its modularity and reliability. Assistive robots, patient-monitoring systems, rehabilitation robots, and medical service robots frequently use ROS 2 for navigation, perception, and user interaction. Service robots in hospitals can use ROS 2 for transportation of medicines, sterilization tasks, and autonomous disinfection using UV systems.

5) Defense, Security, and Disaster Response

Robots used in defense and disaster response often operate in environments with limited connectivity, requiring robust and decentralized communication capabilities. ROS 2's DDS-based middleware is advantageous for field robots used in explosive ordnance disposal (EOD), reconnaissance, hazardous-material detection, and post-disaster search-and-rescue.

Conclusion

ROS 2 represents a significant leap forward in robotics middleware, addressing the limitations of ROS 1 and introducing features suitable for industrial applications. Its distributed architecture, DDS-based communication, real-time capabilities, and security framework make it an ideal choice for modern robotics. While challenges remain in adoption, standardization, and migration, ROS 2 continues to shape the future of robotics research and development. As autonomous systems expand across industries, ROS 2 will play a key role in enabling scalable, reliable, and intelligent robotic solutions.